

UNIT IV

17. (a) Using method of Partial fraction evaluate

$$L^{-1}\left[\frac{1}{s(s+1)s+2}\right]. \quad (6)$$

- (b) Using Convolution theorem find

$$L^{-1}\left[\frac{1}{s(s^2+1)}\right]. \quad (5)$$

Or

18. Using Laplace transform solve

$$\frac{d^2y}{dt^2} + \frac{dy}{dt} = 2e^t, y(0)=1, y'(0)'=2. \quad (11)$$

UNIT V

19. Find Fourier transform for $f(x)=1-x^2, |x|\leq 1$ and

hence deduce that $\int_0^\infty \left[\frac{\sin s - s \cos s}{s^2} \right] \cos \frac{s}{2} ds = \frac{3\pi}{16}.$ (11)

Or

20. Find the Fourier cosine and sine transform for

$$f(x) = \frac{e^{-ax}}{x}, a > 0, x > 0. \quad (11)$$

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B.Tech. DEGREE EXAMINATION,
APRIL 2021.

Second Semester

Common to All Branches

MATHEMATICS – II

(From 2013-14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

- Find the eigen values of the matrix $A = \begin{bmatrix} 5 & 3 \\ 1 & 3 \end{bmatrix}$
- Find the sum and product of Eigen values of the matrix $A = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 2 & 4 \\ 1 & 2 & 7 \end{bmatrix}.$
- Find $\text{Curl } \vec{F}$ if $\vec{F} = (6xy + z^3)\hat{i} + (3x^2 - z)\hat{j} + (3xz^2 - y)\hat{k}.$
- State Gauss Divergence theorem.
- Find Laplace transform for $f(t) = \sin^2 2t.$

6. State and prove change of scale property of Laplace transform.

7. State Convolution theorem on Laplace Transform.

8. Find $L^{-1}\left[\frac{1}{(s+2)^2+1}\right]$.

9. State Fourier integral theorem.

10. Find Fourier sine transform for $f(x) = e^{-ax}$, $a > 0$.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL the Units, choosing ONE full question from each Unit.

UNIT I

11. Verify Cayley Hamilton theorem for the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 4 \\ 3 & 1 & -1 \end{bmatrix} \text{ and hence Find } A^{-1}. \quad (11)$$

Or

12. Reduce the quadratic form $3x^2 + 3y^2 + 3z^2 - 2xy - 2yz + 2zx$ to its Canonical form through an orthogonal transformation. (11)

UNIT II

13. Verify Gauss Divergence theorem for $\vec{F} = x^2\hat{i} + y^2\hat{j} + z^2\hat{k}$ taken over the cube bounded by the lines $x = 0, y = 0, z = 0, x = 1, y = 1, z = 1$. (11)

Or

14. Verify Stoke's theorem for $\vec{A} = (x^2 - y^2)\hat{i} + 2xy\hat{j}$ taken over the rectangular region bounded by lines $x = 0, y = 0, x = a, y = b$. (11)

UNIT III

15. Find the Laplace transform of the rectangular wave given by $f(t) = \begin{cases} 1, & 0 < t < b \\ -1, & b < t < 2b \end{cases}$ with $f(t+2b) = f(t)$. (11)

Or

16. (a) Find $L = \left[\frac{\sin at}{t} \right]$, Hence show that

$$\int_0^{\infty} \frac{\sin t}{t} dt = \frac{\pi}{2}. \quad (6)$$

(b) Verify the initial value theorem on Laplace transform for the function

$$f(t) = 1 + e^{-t}(\sin t + \cos t). \quad (5)$$